Complex numbers

1 What type of number can be written in the form $a + bi$?

2 Complete the following statement:
An imaginary number is a complex number that can be written in the form $\mathbb{I}$.

3 Consider the imaginary number i.

a What is the number i equal to?

b What does i^2 equal?

4 Consider the complex number $-i\sqrt{10}$.

a Write the real part.

b Write the imaginary part.

c State the type of number this is.

5 Write down the complex number that has:

a A real part of 0 and an imaginary part of $\sqrt{5}$.

b A real part of $\sqrt{14}$ and an imaginary part of 0.

Operations with complex numbers

6 Rewrite $9 + \sqrt{-4}$ in the form $a + bi$.

7 Laura wants to evaluate $-\sqrt{-16}$ and writes out the following working out:

$$\begin{aligned} -\sqrt{-16} &= -(-4) \\ &= 4 \end{aligned}$$

Is this working out correct? If not, write the correct answer.

8 Consider the statement:

$$\frac{6 \pm 36i}{2} = 3 \pm 36i$$

Is this statement correct? If not, simplify the statement correctly.

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9 Express the following in terms of i:

a $-\sqrt{-100}$ **b** $\sqrt{-18}$ **c** $9-\sqrt{-64}$ **d** $2+\sqrt{-25}$

10 Express $i\sqrt{-64}$ without a negative radicand.

11 Evaluate:

a $\sqrt{-19}\times\sqrt{-19}$ **b** $-\sqrt{-5}\times\sqrt{-5}$ **c** $\dfrac{\sqrt{-10}}{\sqrt{-2}}$ **d** $\dfrac{\sqrt{-5}}{\sqrt{-45}}$

12 Evaluate:

a $(3+6i)+(7+3i)$ **b** $(9+2i)-(6-7i)$

c $(-1+5i)-2$ **d** $(6-8i)-3i$

e $(6.5+7.9i)+(-9.8-5.3i).$ **f** $(-4+3i)+(-1+5i)+(-6+8i)$

g $(-6-i)-(7-5i)-(-2+9i)$ **h** $(2+3i)-(-6+8i)+(4-i)$

i $\left(\dfrac{1}{2}-\dfrac{i}{4}\right)+\left(\dfrac{2}{3}+\dfrac{i}{5}\right)$

13 Evaluate:

a $-6\,(3-5i)$ **b** $7i\,(3-2i)$

c $\sqrt{10}i\left(8+\sqrt{10}i\right)$ **d** $(5-7i)\,(8+8i)$

e $(1+8i)\,(1-8i)$ **f** $(2+5i)\,(5i-2)$

g $(2-5i)^2$ **h** $\left(\dfrac{4}{5}+\dfrac{3}{5}i\right)\left(\dfrac{1}{4}+\dfrac{3}{4}i\right)$

i $2i\,(3-i)^2$ **j** $-2i\,(4-3i)^2$

14 Consider the imaginary number i.

a Evaluate:

i i^2 **ii** i^4 **iii** i^6 **iv** i^8

b Consider the expression i^n. State whether the following values of n would result in i^n being equal to 1.

i 16 **ii** 22 **iii** 4 **iv** 14

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15 Simplify:

a $2i^7$ **b** vi^{10} **c** $4i^{10}$ **d** $10i^8$

e $(3i)^3$ **f** $(2i)^4$ **g** $5i^2 + 7i^3 - 8i^4$ **h** $8i^3 - 7i^4 + 6i^5$

Complex conjugates

16 The numbers $3 + 4i$ and $3 - 4i$, which differ only in the sign of the imaginary parts, are called what?

17 The product of a complex number and its conjugate is always what type of number?

18 Multiply each of the following numbers by its conjugate:

a $2 + 4i$ **b** $-6 - \sqrt{3}i$ **c** $1 + 5i$ **d** $-4 + 3i$

e $-4 - i$ **f** $\sqrt{2} + i$ **g** $4 - \sqrt{5}i$ **h** $\sqrt{2} + \sqrt{3}i$

19 Evaluate:

a $\dfrac{4 - 6i}{2}$ **b** $\dfrac{5 + 2i}{i}$ **c** $\dfrac{-5}{6i}$ **d** $\dfrac{-5 + 2i}{i}$

e $\dfrac{4 + 6i}{1 + i}$ **f** $\dfrac{7 + 4i}{2 + 3i}$ **g** $\dfrac{4 + 7i}{2 + i}$ **h** $\dfrac{2 - 3i}{2 + 3i}$

i $\dfrac{2 - i}{1 + 4i}$ **j** $\dfrac{2 + 2i}{2 - i}$

20 Consider the expression $5 + \sqrt{6}i$:

a Find the sum of the expression and its conjugate.

b Find the product of the expression and its conjugate.

21 Evaluate:

a $\dfrac{3}{1 + i} + \dfrac{2}{4 - i}$ **b** $\dfrac{8}{i} + \dfrac{7}{i^2} - \dfrac{2}{i^3}$

22 Write $-\dfrac{1}{5}i^3\left(\sqrt{-16} - 25\right) - 4i^2$ in simplest $a + bi$ form.

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Application

23 Complex numbers are used in electronics to describe the current in an electric circuit. Ohm's law relates the current in a circuit, I, in amperes, the voltage of the circuit, E, in volts, and the resistance of the circuit, R, in ohms, by the formula $E = IR$.
Using the formula given above, find E, the voltage of a circuit, if $I = 9 - 4i$ amperes and $R = 1 + 6i$ ohms.

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